Small regular subgroups and their translation kernels

A recovery of Sections 5.1–5.2 of Leemans–Vandenschrick

Let $F = \mathbb{F}_q$ have characteristic p. We first recover the rank-two order bound in Section 5.1, with its unipotent correction. For odd p we then recover the reducible and conic rank-three alternatives of Theorems 5.1–5.2. The actual translation modules also give corrected types for the generic affine row of Table 3. We retain the standard string end-intersection criterion, and Dickson's classification of subgroups of $\mathrm{PGL}_2(F)$ in its natural subfield form. The latter is an attributed classical input, not a new classification claimed here.

Two involutions

Theorem 1. *If distinct involutions generate a rank-two subgroup of* $\mathrm{PSL}_3(q)$*, its order divides* $2(q-1)$*,* $2(q+1)$ *or* $4p$*. In characteristic two, the last possibility has rotation order at most four. In odd characteristic a regular unipotent rotation has order* p*.*

Proof. The scalar centre has odd order dividing three. Each projective involution has a unique involutory lift to determinant one, and the product X of the two lifts is conjugate to X^{-1}. Its eigenvalues are therefore $1, \lambda, \lambda^{-1}$. If $\lambda \neq \pm 1$, the matrix is semisimple: either $\lambda \in F$ and its order divides $q-1$, or $\lambda^q = \lambda^{-1}$ and its order divides $q+1$. If all eigenvalues are one, the largest Jordan block has size at most three. Its order divides p for odd p, and four for $p = 2$. In odd characteristic the remaining case has eigenvalues $1, -1, -1$; its order divides $2p$. Projectivization does not shorten any of these orders: if $X^m = \zeta I$, its eigenvalue one forces $\zeta = 1$. The group generated by the two involutions is dihedral of twice the rotation order, giving the assertion.

These are the expected ambient dihedral families, not just numerical bounds. The usual split and nonsplit tori in $\mathrm{PGL}_2(q)$ have normalizers $D_{2(q-1)}$ and $D_{2(q+1)}$; their three-dimensional orthogonal representation embeds in $\mathrm{PSL}_3(q)$ when q is odd. When q is even the block embedding of $\mathrm{SL}_2(q)$ gives the same normalizers. In odd characteristic, $\mathrm{diag}(-1,-1,1)(I + E_{12})$ has order $2p$ and is inverted by $\mathrm{diag}(1,-1,-1)$, giving D_{4p}. In characteristic two, $I + E_{12}$ and $I + E_{23}$ give D_8. Their dihedral subgroups realize the corresponding smaller orders. □

The source's displayed regular unipotent on physical page 11 is incorrectly assigned the bound $2(2,p)$. Its order is p for odd p and four for $p = 2$. The final dihedral order bound remains valid. This is an isomorphism and order classification; no assertion that all embeddings of a given dihedral group are conjugate is needed.

A common block form in odd characteristic

Now suppose p is odd. Put

$$d_1 = \mathrm{diag}(1,-1,-1), \quad d_2 = \mathrm{diag}(-1,1,-1), \quad d_3 = d_1 d_2.$$

The commuting distinct end generators of a rank-three marking generate the diagonal Klein group $D = \{1, d_1, d_2, d_3\}$ after a coordinate change. The middle generator is a homology, with a centre and a complementary axis. Suppose first that the whole group fixes a point or line.

The common eigenspaces of D are exactly the coordinate points and coordinate planes. After duality if necessary, the middle axis therefore contains a coordinate point. Move this point to e_3. The middle lift has the form

$$t = \begin{pmatrix} A & 0 \\ v & -1 \end{pmatrix}, \qquad A = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}, \quad a^2 + bc = 1, \quad vA = v. \tag{1}$$

The group is $G = \langle D, t \rangle$, regardless of which two nonidentity members of D are used as its ordered outer generators. Every such group fixes e_3 as a projective point. Its matrix lifts have bottom eigenvalue ± 1, so they meet the scalar centre trivially. We may work throughout with these actual matrix groups.

Write

$$M(x, y) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ x & y & 1 \end{pmatrix}.$$

A pure block matrix with upper block B and lower scalar $\delta = \det B \in \{1, -1\}$ acts by conjugation on these rows as $(x, y) \mapsto \delta(x, y)B$. This elementary action, together with coordinate sign projections, accounts for the generic cases.

Theorem 2. *Suppose $bc \neq 0$. Normalize*

$$A = \begin{pmatrix} a & -a - 1 \\ a - 1 & -a \end{pmatrix}, \qquad a \neq \pm 1. \tag{2}$$

Then v is either zero or can be normalized to $(1, -1)$. Put $K = \mathbb{F}_p(a)$, choose λ with $\lambda + \lambda^{-1} = 2a$, and set

$$h = |\lambda|, \quad h' = |-\lambda|, \quad m = \operatorname{lcm}(h, 2).$$

If $v = 0$, the generated group is D_{2m}. If $v = (1, -1)$, it is $K^2 \rtimes D_{2m}$ in the displayed faithful two-dimensional action. All three choices of distinct outer generators in D then give full rank-three IP. Up to reversal their types are

$$[h, h'], \qquad [h, 2p], \qquad [h', 2p]. \tag{3}$$

At $a = 0$ the group is $D_{2p} \wr C_2$ and these types reduce to $[4, 4]$ and $[4, 2p]$.

Proof. Diagonal scaling gives (2), and its left $+1$ eigenspace is $\langle (1, -1) \rangle$. Scaling the third coordinate gives the asserted normalization of v. Let $h_A = \operatorname{diag}(A, -1)$. For $v = (1, -1)$, the actual kernel of the upper-block action contains

$$(td_3)^2 = M(2, -2).$$

Conjugating by D and adding or subtracting the results separates the two coordinates. Dividing by two is valid in odd characteristic. Conjugation by t and projection then multiply a coordinate by a. Iteration gives every element of K in each coordinate. Conversely all matrices have entries in K. Thus this kernel is exactly $M(K^2)$. Since $t = M(v)h_A$, the group contains the pure block lift h_A and is the actual semidirect product $M(K^2) \rtimes H$, where $H = \langle D, h_A \rangle$. In the zero case it is just H.

The upper block of $d_1 h_A$ has determinant one and eigenvalues λ, λ^{-1}. They are distinct and not ± 1. Adjoining d_3, whose upper block is $-I$, enlarges its cyclic group to order m: for even h, its half-power is already $-I$, whereas for odd h adjoining $-I$ doubles the order. The remaining reflection inverts this cyclic group. Hence $H = D_{2m}$. In particular its action on K^2 is faithful and no nonidentity rotation fixes a nonzero vector. The trace field is K; equivalently it is the least field in which $\lambda + \lambda^{-1}$ lies. Thus m divides $|K| - 1$ or $|K| + 1$.

For completeness we prove the actual intersections. Put $A_i = \langle d_i, t\rangle$, and let $H_i = \langle d_i, h_A\rangle$ for $i = 1, 2$. With the convention $H^g = g^{-1}Hg$, direct multiplication gives

$$A_i = H_i^{M(c_i)}, \qquad c_1 = (0, (1-a)^{-1}), \quad c_2 = (-(a+1)^{-1}, 0). \tag{4}$$

The translations c_i fix d_i by conjugation and send h_A to t. The vectors differ. Therefore the common elements of these two complements inject into the stabilizer of $c_1 - c_2$ in H. Such a stabilizer has order at most two: a nonidentity rotation fixes no nonzero vector, and two reflections would have such a rotation as their product. Since both ends contain t, their intersection is exactly $\langle t\rangle$.

The third end is

$$A_3 = M(\mathbb{F}_p v) \rtimes \langle d_3, h_A\rangle, \qquad |A_3| = 4p. \tag{5}$$

Indeed its squared rotation gives $M(2v)$, which gives $M(v)$ and hence h_A. Its upper-block image is the Klein group $\langle d_3, h_A\rangle$. Each A_i, $i = 1, 2$, injects into that block action. If an element of $A_i \cap A_3$ had image d_3, equation (4) would force its translation to be $-2c_i$. This is not in the line Kv, and hence not in the kernel line of (5). The image $d_3 h_A$ is also excluded by multiplication by the common element t. Only the identity and h_A images remain, so $A_i \cap A_3 = \langle t\rangle$. The rank-three end criterion now proves full IP for every outer pair.

The rotations $d_1 t, d_2 t$ have orders h, h': their upper blocks have no eigenvalue one, so the affine displacement contributes nothing to their order. Equation (5) gives $|d_3 t| = 2p$. This proves (3). If $a = 0$, the block group is the signed permutation group on two coordinates. Its sign changes, together with $\mathbb{F}_p^2$, form two independent D_{2p} factors, exchanged by its coordinate swap. This is exactly the asserted wreath product. □

A dihedral group admits a rank-three string C-marking only in the degenerate case D_{4k} with k odd, of type $[k, 2]$ or its reverse. Here is the short argument needed for the zero case above. If all three marks are reflections, the commuting ends differ by the central involution; their common nontrivial squared rotation violates IP unless an adjacent product has order two. In that remaining situation a middle reflection commuting with an end is either that end or the other end, again violating independence. Thus the central involution is an end mark. The other two marks generate a dihedral subgroup intersecting it trivially, and the whole group is their direct product with C_2. This is dihedral exactly when their rotation order k is odd. The middle mark cannot be central, since then the whole group would be an elementary abelian group of rank at most two.

The triangular specializations

If $b = c = 0$ in (1), either $t \in D$, which fails IP, or the group is D_{4p}. If exactly one of b, c vanishes, interchanging the first two coordinates makes $a = 1$. Scaling the remaining entry and, when nonzero, the bottom row gives one of the following matrices, or the same D_{4p} case:

$$t_P = \begin{pmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix}, \qquad t_H = \begin{pmatrix} 1 & -2 & 0 \\ 0 & -1 & 0 \\ 1 & -1 & -1 \end{pmatrix}. \tag{6}$$

For example when $b = 0, c \neq 0$, the equation $vA = v$ makes $v = (v_1, 0)$; when $c = 0, b \neq 0$, it makes $v_2 = bv_1/2$. This proves completeness of (6), including the cases with zero bottom row.

Theorem 3. *The group $\langle D, t_P\rangle$ is $D_{2p} \times D_{2p}$. Its two valid nondegenerate outer pairs have type $[p, 2p]$, up to reversal; the remaining pair fails IP. The group $\langle D, t_H\rangle$ is $\mathrm{Heis}(p) \rtimes C_2^2$. All three outer pairs have IP, with types $[p, 2p]$, $[p, 2p]$, $[2p, 2p]$, up to reversal.*

Proof. In the product case write a translation as $n(x,y) = I + xE_{21} + yE_{31}$. Then $t_P = n(1,1)d_1$. The diagonal signs separate its two prime-field coordinates: d_2 inverts the first, d_3 the second, and d_1 both. Thus the actual group is the direct product claimed. The end $A_1 = \langle d_1, t_P \rangle$ has normal forms $n(k,k)d_1^\epsilon$. In $A_2 = \langle d_2, t_P \rangle$ the coordinate x is arbitrary and y is zero or one according as its diagonal component fixes or negates the second coordinate. For A_3 the analogous condition holds on x. These descriptions follow by squaring $d_2 t_P, d_3 t_P$, and give $A_1 \cap A_2 = A_1 \cap A_3 = \langle t_P \rangle$, whereas $|A_2 \cap A_3| = 4$. The orders of $d_i t_P$ are respectively $p, 2p, 2p$.

For the second case use

$$U(x,y,z) = I + xE_{12} + yE_{31} + zE_{32}, \qquad U(x,y,z)U(X,Y,Z) = U(x+X, y+Y, z+Z+yX).$$

We have $t_H = U(2,1,1)d_1$. Conjugation by d_2 changes this unipotent factor to $U(-2,1,-1)$. Their commutator has central coordinate -4, so it generates the full prime-field central root group. Their product then supplies the second root group, and the first follows by division. This proves the actual full Heisenberg kernel and its diagonal complement D.

Put $A_i = \langle d_i, t_H \rangle$. The first end has forms $U(2k,k,k^2)d_1^\epsilon$. The kernels of the diagonal projections of the other two ends are respectively

$$N_2 = \{U(x,0,x/2) : x \in \mathbb{F}_p\}, \qquad N_3 = \{U(0,y,-y) : y \in \mathbb{F}_p\}.$$

These follow by squaring $d_2 t_H, d_3 t_H$; each end has order $4p$. The first kernel meets either of them trivially, giving $A_1 \cap A_2 = A_1 \cap A_3 = \langle t_H \rangle$. For the remaining intersection the kernels are disjoint. At diagonal component d_2, the possible translations in A_2, A_3 are respectively $U(x,0,x/2)$ and $U(2,t,t)$; equality would force $x = 2, t = 0$ and then $1 = 0$. At component d_3 they are $U(x-2,1,x/2-1)$ and $U(0,y,-y)$; equality would give $x = 2, y = 1$ and then $0 = -1$. Thus only the identity and d_1 components remain, with unique common lifts $1, t_H$. This proves the final intersection. The rotation orders are $p, 2p, 2p$, and the end criterion completes the proof. □

The irreducible conic alternative

Theorem 4. *If an odd-characteristic rank-three string C-group in* $\mathrm{PSL}_3(F)$ *fixes no projective point or line, it preserves a nonsingular conic. Its group is one of* $S_4, A_5, \mathrm{PSL}_2(K), \mathrm{PGL}_2(K)$, *in the natural three-dimensional orthogonal representation over a subfield* K.

Proof. With the diagonal outer Klein group, write the middle homology as $t = -I + 2uf$, where $f(u) = 1$. Every coordinate of u and f is nonzero: a zero coordinate of u would preserve the corresponding coordinate plane, and a zero coordinate of f the coordinate point. The diagonal quadratic form

$$Q = \mathrm{diag}(f_1/u_1, f_2/u_2, f_3/u_3)$$

is nonsingular, has $Q(u,u) = 1$, and is preserved by D and t. Its determinant-one orthogonal group meets the scalar centre trivially, and is isomorphic to $\mathrm{PGL}_2(F)$. Apply the retained Dickson theorem. Semisimple cyclic and dihedral subgroups preserve the torus axis in this orthogonal representation. The remaining cyclic or dihedral cases lie in a point stabilizer in the projective-line action, which preserves its corresponding isotropic point. The same holds for the Borel-type subgroups in Dickson's list. These are excluded by hypothesis. An A_4 cannot be generated by involutions, whose generated subgroup is only V_4. The stated alternatives remain. □

For S_4 the nondegenerate types are $[3,3]$ and $[3,4]$, up to reversal. Both adjacent orders cannot be four: the involution types in S_4 would give only a subgroup D_8. For A_5 the possible types are $[3,5]$ and $[5,5]$: a $[3,3]$ marking would be a quotient of the Coxeter group S_4. The

usual tetrahedral and projective cubical and icosahedral markings supply these small possibilities. The wider rank-three marking classifications within the projective-line groups remain separately attributed inputs; we do not repeat them here.

The source table and the corrected affine row

Theorems 2–4 recover the group alternatives of Theorems 5.1–5.2. They also give every non-degenerate affine type in Table 3, with one necessary correction. The generic row in that table prints only $[k, 4]$ and $[k, k]$. Its literal middle matrix (2) with bottom row $(1, -1, -1)$ instead gives the three types (3). For a concrete instance take $F = \mathbb{F}_5$ and $a = 2$. Then $h = 3, h' = 6$, $K = \mathbb{F}_5$, $m = 6$, and the group has order 300. The three actual C-markings have types $[3, 6]$, $[3, 10]$, $[6, 10]$. Their full subset intersections are checked in the supplied finite control; the uniform proof above gives them independently of that computation. This is a correction to a type row, not a refutation of the subgroup theorem or of the paper's principal chiral constructions and rank bound.

The centre–axis normalization and original matrices are due to Leemans–Vandenschrick. The proof here replaces the separate geometric case calculations by one block action, its translation kernel and actual complement intersections. No numerical compression or new conjecture resolution is asserted by this component.